

Dahan River Watershed Five Waterways Digital Walking Tour System

System Plan

Course	Web GIS Application Development
Group	Group J
System Name	Dahan River Watershed Five Waterways Digital Walking Tour System
Technology Stack	OpenLayers 9 · HTML5 / CSS3 / JavaScript
Revision	Revision 1
Date	June 2026

Revision History

Revision	Date	Author	Description
1	June 2026	Group J	Initial release — system plan, functional specification, UI element descriptions, and diagrams

1. System Specification

1.1 Requirement Definition

The Dahan River Watershed Five Waterways Digital Walking Tour System is a web-based GIS application built entirely with open-source software and open standards. The system provides an interactive digital experience for exploring the cultural, ecological, and historical significance of five major tributaries of the Dahan River, located across New Taipei City and Taoyuan City in northern Taiwan.

System requirements are as follows:

- Open-source software only: OpenLayers 9 as the GIS mapping library; no proprietary dependencies.
- OGC-compliant basemap tile services (OSM / Esri WMTS / OpenTopoMap).
- WGS84 (EPSG:4326) as the native coordinate reference system; displayed in Web Mercator (EPSG:3857).
- Multi-page web application accessible via standard desktop browsers without installation.
- Spatial database structured as JavaScript objects conforming to feature class conventions (Polyline, Point, Non-spatial Table).
- Responsive design supporting desktop and mobile viewport widths.

1.2 Development Environment

Target web browsers:

- Google Chrome (primary)
- Microsoft Edge
- Apple Safari

Development tools and technologies:

- Coding: HTML5, CSS3, Vanilla JavaScript (ES6+)
- GIS map library: OpenLayers 9 (open-source, CDN-delivered)
- Basemap tile services: OpenStreetMap (OSM), Esri World Imagery WMTS, OpenTopoMap
- Fonts: Google Fonts — Noto Serif TC, Noto Sans TC
- Development server: XAMPP (local) + ngrok (public demo tunnel)

Data standards:

- Coordinate system: WGS84 / EPSG:4326 (storage); EPSG:3857 (display via `ol.proj.fromLonLat()`)
- Spatial data format: JavaScript object literals structured as GeoJSON-equivalent feature classes
- Timeline data: Non-spatial event table (year, stream reference, title, description)

1.3 System Scope

The system comprises four integrated sub-pages that together form a complete Web GIS application:

Sub-page	Path	Primary Function
Home	index.html	Landing page with hero section; waterway quick-access badges; stream info modal popup
Interactive Map	map.html	Main GIS map view with all operational layers, tools, and guided walking tour mode
Waterway History	history.html	Chronological timeline of historical events filterable by waterway
Waterway Overview	about.html	Statistical summary and detailed data cards for all five waterways

The shared data layer (shared.js) serves as the system spatial database, providing STREAMS, STREAM_PATHS, TIMELINE, STATS, and GEODATABASE objects to all sub-pages.

The following diagram shows the system architecture and the relationships between sub-pages and the shared data layer:

2. System Development Plan

The system development plan is structured into five functional modules corresponding to the five Web GIS requirements. The following diagram shows the overall system tree structure:



Figure 1: System Tree Structure Diagram

2.1 Web Application Module

System Structure

The application is a static multi-page website. All pages share a common navigation bar, CSS variable design system, and the shared.js data layer. No server-side rendering is used.

File	Role
index.html	Home page — hero banner, stream badges, modal popup
map.html	Interactive GIS map — all tools and layers
history.html	Timeline page — filterable by waterway
about.html	Overview page — stats and waterway data cards
shared.js	Shared spatial database — STREAMS, STREAM_PATHS, TIMELINE, GEODATABASE
ol.js / ol.css	OpenLayers 9 library (local copy)

Conceptual UI Elements

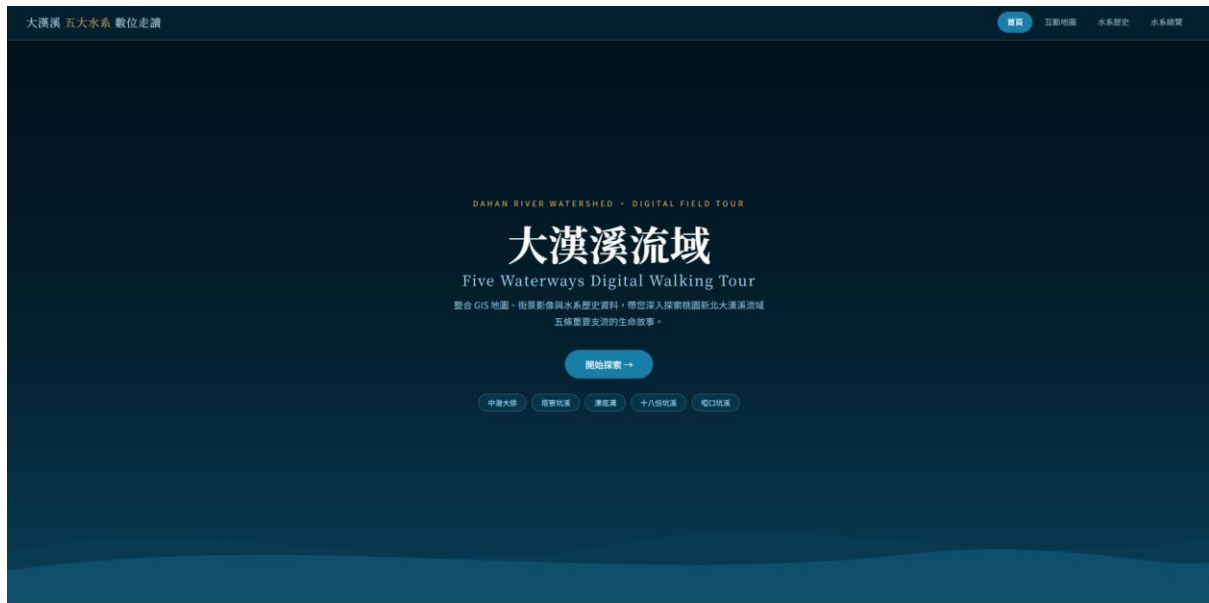


Figure 2: Home Page (index.html) — Conceptual UI Elements

Main Functions

- Unified navigation bar with active-state highlighting across all four pages.
- Consistent dark-blue design system using CSS custom properties (--deep, --accent, --pale, --gold).
- Stream modal popup on the home page — clicking any waterway badge opens a detailed overlay with description, tags, history, and a Google Maps external link.
- Responsive layout supporting viewport widths from 320px (mobile) to 1920px (desktop).

2.2 Digital Basemaps Module

System Structure

Three tile-based basemap layers are pre-loaded as OpenLayers Tile layer instances. Only one is visible at a time; switching is instantaneous with no network re-initialization.

Basemap ID	Source	Tile URL Pattern	Use Case
OSM (default)	OpenStreetMap	tile.openstreetmap.org/{z}/{x}/{y}.png	General street reference
Satellite	Esri World Imagery	server.arcgisonline.com/.../tile/{z}/{y}/{x}	Aerial/satellite imagery
Topographic	OpenTopoMap	{a-c}.tile.opentopomap.org/{z}/{x}/{y}.png	Terrain and elevation context

Conceptual UI Elements

A basemap selector panel in the top-right corner of the map view displays three pill-shaped buttons: 🗺️ Street Map, 🌐 Satellite, ▲ Topographic. The active basemap is highlighted in the system accent color (#1a7fa8).

Element ID	Label	Function
bm-osm	🗺️ Street Map (街道圖)	Activates OpenStreetMap basemap; default on page load
bm-sat	🌐 Satellite (衛星圖)	Activates Esri World Imagery WMTS layer
bm-topo	▲ Topographic (地形圖)	Activates OpenTopoMap terrain layer

2.3 Operational Layers Module

System Structure

Three independent vector layers are rendered on top of the basemap. Each layer is controlled by a separate checkbox in the Layer Control panel and can be toggled without affecting other layers.

Layer Name	Geometry Type	Source Data	Rendering Style
Waterway Lines (河川水系線)	Polyline / MultiPolyline	STREAM_PATHS[] in shared.js	Colored stroke (5px, rounded), unique color per waterway
Tour Marker Points (導覽標記點)	Point	STREAMS[i].pois[] in shared.js	Filled circle (r=7px, white stroke, waterway color fill)
Stream Name Labels (水系名稱標籤)	Point (Text)	STREAM_PATHS midpoint	Bold 11px label, waterway color, dark outline stroke

Layer Control UI Elements

Element ID	Layer Controlled	Default State
lyr-streams	Waterway Lines (河川水系線)	Checked (visible)
lyr-labels	Stream Name Labels (水系名稱標籤)	Checked (visible)
lyr-pois	Tour Marker Points (導覽標記點)	Checked (visible)

2.4 Tasks and Tools Module

Conceptual UI Elements

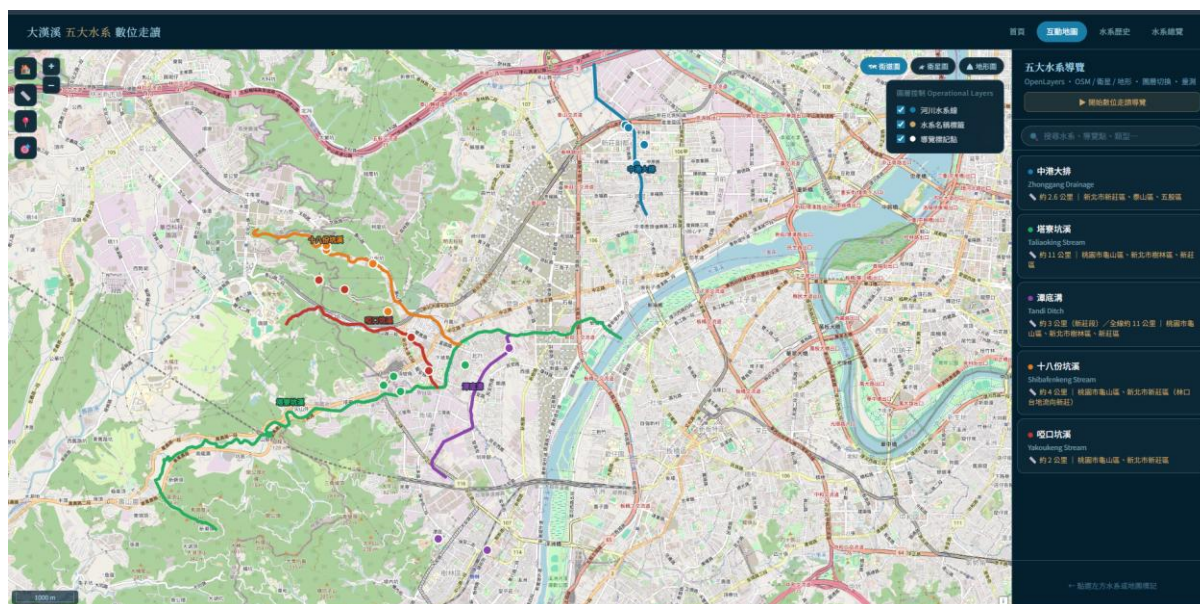


Figure 3: Interactive Map Page (map.html) — Conceptual UI Elements

Map View (M0) — Main Functions

The main map occupies the full left area of the screen. The sidebar (right, 320px) shows the waterway list, search bar, and detail panel. On mobile, the layout stacks vertically.

Element ID	Label	Function
btn-home	Reset View	Resets map to initial center (121.42°E, 25.02°N) at zoom 11 with 800ms animation
btn-measure	Measure	Toggles distance measurement mode; draws a dashed line; displays length in meters or km using <code>ol.sphere.getLength()</code>
btn-poi	POI Toggle	Toggles visibility of all tour marker points; syncs with layer panel checkbox
btn-locate	My Location	Activates Geolocation API; places an animated dot at user position; calculates nearest tour POI distance
bm-osm / bm-sat / bm-topo	Basemap Buttons	Switch between three basemap tile layers (see 2.2)
lyr-streams / lyr-labels / lyr-pois	Layer Checkboxes	Toggle individual operational layers on/off (see 2.3)

POI Popup (M0-P)

Clicking any tour marker point opens a popup window anchored to the marker coordinate. The popup uses pixel-coordinate positioning (`getPixelFromCoordinate`) and follows the map during pan/zoom via the postrender event.

Element	Content
Stream badge	Waterway name in waterway color — e.g., Tandi Ditch (purple)
POI title	Name of the tour marker point
Type label	Category of the POI (e.g., Historical Site, Park Trail, Religious Culture)
 Street View	Opens Google Street View link in new tab
 Map	Opens Google Maps search for the POI location in new tab
✕ Closer	Closes the popup

Search Function (M0-S)

A search bar in the sidebar allows querying both waterways and POIs by name, type, basin, or tag keyword. Results are shown in a dropdown with a colored stream badge indicating which waterway each result belongs to.

Element ID	Label	Function
s-input	Search Input	Text input; triggers live search on each keystroke; placeholder: Search waterways, POIs, types...
s-clear	✕ Clear	Clears the search input and closes the dropdown
s-drop	Results Dropdown	Shows matched waterways (section: Waterway) and POIs (section: Tour Points); each item has a stream badge, name, and type

Digital Walking Tour Mode (M0-T)

Pressing ► Start Walking Tour activates a guided tour that sequentially visits all 15 tour points across the five waterways. Each waterway is preceded by a full-screen introduction card showing the stream name, description, tags, and issue notes.

Element	Function
tour-start-btn	Toggles tour mode on/off; changes label to ✕ End Tour when active
Stream Intro Card	Full-screen overlay before each waterway; contains name, EN name, length, basin, description, tags, issues, and Enter/Skip buttons
Tour Control Bar	Bottom panel showing current stop title, description, progress dots, and Prev / Next / End controls
Tour Complete Screen	Shown after all stops are visited; displays statistics and a link to the History page

2.5 Waterway History and Overview Module

Conceptual UI Elements



Figure 4: Waterway History Page (history.html) — Conceptual UI Elements



Figure 5: Waterway Overview Page (about.html) — Conceptual UI Elements

2.6 Geodatabase Module

Spatial Database Structure

The system spatial database is implemented as structured JavaScript objects in shared.js, following geodatabase feature class conventions. All coordinate data uses WGS84 (EPSG:4326); the map renders in Web Mercator (EPSG:3857) via ol.proj.fromLonLat().

Feature Class	Geometry	Record	Key Fields	Description
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	Type	Count		
streams_line	Polyline / MultiPolyline	5	stream_id, name, name_en, length_km, basin, color_hex	Waterway centerline geometry; stored as ordered [lng, lat] coordinate arrays in STREAM_PATHS[]
pois_point	Point	15	poi_id, stream_id, name, type, lat, lng, sv_url, img_url, desc	Tour marker point features with street-view links; embedded in STREAMS[i].pois[]
timeline_events	Non-spatial Table	12	year, stream_id, color_hex, title, text	Historical event records stored in TIMELINE[]; used by the history page timeline visualization

Waterway Feature Class (streams_line)

ID	Name	English Name	Length	Basin / Administrative Area
0	中港大排	Zhonggang Drainage	~12.5 km	Xinzhuang Dist., New Taipei City; Zhongli Dist., Taoyuan City
1	塔寮坑溪	Taliaoking Stream	~8.3 km	Guishan Dist., Taoyuan; Shulin & Xinzhuang Dist., New Taipei
2	潭底溝	Tandi Ditch	~5.7 km	Yingge Dist., New Taipei / Daxi Dist., Taoyuan (border)
3	十八份坑溪	Shibafenkeng Stream	~6.1 km	Xinzhuang Dist., New Taipei City (Linkou Plateau to Xinzhuang)
4	啞口坑溪	Yakoukeng Stream	~4.8 km	Xinzhuang Dist., New Taipei City (tributary of Taliaoking Stream)

Tour Point Feature Class (pois_point)

Stream	POI Name	Type
Zhonggang Drainage	Zhonggang Drainage Waterfront Trail	Park Trail
Zhonggang Drainage	Rainbow Plaza	Waterfront Facility
Zhonggang Drainage	Drainage Outlet Gate	Hydraulic Structure
Taliaoking Stream	Taliaoking Stream Upper Natural Section	Natural Attraction
Taliaoking Stream	Old Brick Kiln Ruins Area	Historical Ruin
Taliaoking Stream	Huilong Settlement Section	Historic Settlement
Tandi Ditch	Tandi Ditch Stone-Stacked Revetment Section	Cultural Heritage
Tandi Ditch	Longan Bridge	Landmark Bridge
Tandi Ditch	Tandi Ditch Agricultural Irrigation Intake	Hydraulic Structure

Shibafenkeng Stream	Shibafenkeng Stream Natural Valley	Natural Attraction
Shibafenkeng Stream	Old Camphor Processing Ruins Area	Historical Ruin
Shibafenkeng Stream	Waterworks Memorial Park	Cultural Site
Yakoukeng Stream	Yakou Bridge	Landmark Bridge
Yakoukeng Stream	Shuangfeng Park	Park & Green Space
Yakoukeng Stream	Fuzheng Temple	Religious Culture

3. Waterway Historical Timeline

The following historical events are stored in the TIMELINE non-spatial table and displayed on the history.html page. Users can filter events by waterway using the filter bar.

Year	Waterway	Event Title	Description
1684	Zhonggang Drainage	Early Qing Dynasty Irrigation Development	Zhangzhou and Quanzhou settlers entered the Taoyuan Plain; irrigation ditches were dug, serving as the precursor to the Zhonggang Drainage system.
1760	Shibafenkeng Stream	Shibafenken Land Reclamation	Large-scale reclamation during the Qianlong era; land was divided into shares (fen), and the Shibafenkeng Stream basin agricultural development was completed.
1800	Tandi Ditch	Tandi Ditch Agricultural Irrigation	The ditch served as an agricultural water source during the Jiaqing era; stone-stacked revetment construction techniques established, some of which survive today.
1895	Taliaoking Stream	Japanese Colonial Era Brick Kilns Established	Multiple brick kilns were set up along Taliaoking Stream in early Japanese colonial period, supplying building materials for northern Taiwan; stream water used for kiln cooling.
1908	Shibafenkeng Stream	Camphor Processing Industry Peak	Camphor production in the Daxi area reached its peak; large-scale logging of camphor trees in the Shibafenkeng upper watershed for camphor oil extraction.
1928	Shibafenkeng Stream	Xinzhuang Waterworks Inaugurated	The Xinzhuang waterworks system was built near the Shibafenkeng Stream water source, supplying water to approximately 50,000 residents; a major modernization milestone.
1930	Yakoukeng Stream	Yakoukeng Forest Control Station Established	Japanese authorities set up a control station at the Yakou pass to regulate forest resources; Yakoukeng Stream became the boundary between Han settlers and indigenous territories.
1960	Zhonggang Drainage	Zhongli Industrialization Impact	Post-war industrial zones established in Zhongli; large amounts of industrial wastewater discharged into the Zhonggang Drainage, causing severe water quality deterioration.
1980	Tandi Ditch	Ditch Culverting Era	Rapid urbanization led to multiple sections of Tandi Ditch being covered and converted to underground culverts; surface waterways gradually disappeared from the landscape.
2010	Zhonggang Drainage	Waterfront Corridor Renovation Plan	New Taipei City launched an environmental improvement plan for the Zhonggang Drainage, adding waterfront trails and ecological green belts.

2018	Taliaoking Stream	Taliaoking Stream Ecological Survey	Environmental groups documented multiple species of migratory birds and native fish; natural bank conservation advocacy initiated.
2023	Shibafenkeng / Yakoukeng	Daxi Waterway Cultural Route Planning	Daxi District launched a waterway cultural route plan, integrating the history and culture of Shibafenkeng and Yakoukeng streams into tourism routes.

4. System Error Messages and Status

Message ID	Context	Message Text	Trigger Condition
M0-M01	Geolocation	Location failed. Please ensure browser location permission is enabled.	Geolocation API returns error or permission denied by user
M0-M02	Measure Tool	Click to set a start point, continue clicking, double-click to finish.	User activates the measure tool; shown as hint in the measure panel
M0-M03	Search	No results found for "{keyword}"	Search query matches no streams or POIs in the database
M0-M04	Basemap Load	(No explicit message — tile load failure shows blank tiles)	Network error when loading tile layers; logged to browser console only
M0-M05	Tour Complete	Walking Tour Complete!	User finishes all tour stops across all five waterways

5. System Architecture Summary

The following table summarizes how the system fulfills each of the five Web GIS application requirements:

Requirement	Implementation	Status
1. Web Application	Four-page HTML5/CSS3/JS static website; unified nav bar, shared data layer (shared.js), responsive design	✓ Implemented
2. Digital Basemaps	Three switchable tile basemaps: OSM, Esri World Imagery (WMTS), OpenTopoMap; one-click toggle buttons in top-right panel	✓ Implemented
3. Operational Layers	Three independent vector layers (Polyline stream lines, Point POIs, Text labels); each togglable via layer control panel	✓ Implemented
4. Tasks and Tools	Distance measurement (ol.sphere.getLength), basemap switching, layer control, POI popup with street view, keyword search, geolocation, digital walking tour mode	✓ Implemented
5. Geodatabases	shared.js implements streams_line (Polyline×5), pois_point (Point×15), timeline_events (Table×12); WGS84 EPSG:4326 coordinates; rendered in EPSG:3857	✓ Implemented